

WATER QUALITY ANALYSIS REPORT

(Python & Power BI Project)

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Project Type: Data Analytics – Environmental Study
Tools Used: Python, Pandas, Scikit-learn, Matplotlib, Power BI
Dataset Coverage: 2,610 observations | 10 locations | 2020–2024
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Abstract

This project presents an in-depth analysis of water quality across multiple monitoring locations over a five-year period. Using key physical, chemical, and biological indicators, a Water Quality Index (WQI) is developed to assess overall conditions.

Advanced analytics techniques including machine learning and anomaly detection are applied to identify pollution patterns and predict water quality categories. The results are further visualized through an interactive Power BI dashboard to support decision-making.

The study highlights seasonal variations, pollution hotspots, and key environmental risks affecting urban water systems.

Objectives

- 1 Evaluate water quality across multiple monitoring stations
- 2 Develop a composite Water Quality Index (WQI)
- 3 Analyze seasonal and geographical trends
- 4 Identify high-risk pollution zones
- 5 Apply machine learning for classification
- 6 Detect unusual or abnormal readings
- 7 Present insights through interactive dashboards

Dataset Overview

The dataset represents simulated yet realistic water quality observations collected across 10 monitoring stations over a period of five years.

- 1 Total records: 2,610
- 2 Locations: 10 stations
- 3 Time period: 2020 to 2024

Parameters analyzed include:

- 1 Physical: pH, temperature, turbidity, total dissolved solids
- 2 Chemical: dissolved oxygen, nitrates, phosphates
- 3 Biological: BOD, COD, coliform levels
- 4 Heavy metals: lead, arsenic, chromium

Methodology

The analysis follows a structured data science workflow:

- 1 Data Preparation: Missing values handled using statistical imputation; time features derived
- 2 Index Calculation: WQI computed using weighted normalized parameters
- 3 Data Analysis: Patterns and trends identified using exploratory analysis
- 4 Machine Learning: Random Forest model used for classification
- 5 Anomaly Detection: Isolation Forest used to detect unusual observations
- 6 Visualization: Charts and dashboards for interpretation

Key Analysis Insights

- 1 Industrial locations show lower water quality
- 2 Reservoirs maintain better quality
- 3 Strong link between organic pollution and microbial contamination
- 4 Dissolved oxygen inversely related to pollution
- 5 Peak pollution occurs during summer

Statistical Summary

- 1 pH: ~7.0
- 2 Dissolved Oxygen: ~6.6 mg/L
- 3 BOD: ~4.0 mg/L
- 4 TDS: ~500 mg/L
- 5 WQI: ~75

These values indicate varying pollution levels across locations.

Machine Learning Results

A Random Forest model achieved approximately 94% accuracy.

- 1 Key factors: BOD, Dissolved Oxygen, Coliform levels

Major Findings

- 1 Industrial zones are most polluted
- 2 Reservoirs are cleanest
- 3 Summer has highest pollution
- 4 Heavy metal contamination present
- 5 Pollution linked to seasonal and industrial activity

Recommendations

- 1 Control industrial discharge
- 2 Increase monitoring frequency
- 3 Use real-time sensors
- 4 Improve treatment infrastructure
- 5 Conduct environmental audits

Dashboard Overview

- 1 Station-wise tracking
- 2 Seasonal trends
- 3 Risk categorization
- 4 KPI indicators

License

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